# NBDesigner Universal Handshake Blueprint

Engineering Specification: Handshaking Any Client-Side Product Designer with the Native NBDesigner Backend

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|---|---|
| **Target Backend:** NBDesigner v2.8.5+ (WooCommerce) | **Architecture Pattern:** 10-Artifact Decoupled Handshake |
| **Integration Layer:** Zero-Canvas Headless Handshake | **Print Fidelity:** 300 DPI Transparent Raster + TCPDF Vector |

## 1. Executive Architectural Blueprint

This specification establishes the architectural contract for handshaking **any independent client-side product customizer** (such as form-based editors, 3D Canvas visualizers, or custom SVG UI engines) with the native **NBDesigner** (web-to-print-online-designer) backend. Under this architecture, the external editor acts as an autonomous competitor or alternative layout on the storefront. Once the customer completes their personalization, the editor transmits a clean payload into NBDesigner's backend file system and WooCommerce order metadata.

**The Core Advantage:** Developers need only focus on building the front-end user experience. Once the order is placed, store administrators log into the standard WooCommerce Order management screen and utilize NBDesigner's native tools: **View Detail** (inspecting text fonts, sizes, and colors), **Download PNG (300 DPI)**, **Download SVG**, **Create Vector PDF** (via TCPDF), and **Download All ZIP**, with zero modification to NBDesigner core files.

## 2. The 10-Artifact Handshake File Convention

NBDesigner stores design assets in wp-content/uploads/nbdesigner/designs/{nbd_item_key}/. To satisfy all native export hooks without triggering PHP warnings or aborts, the client-side bridge must create the following files:

| Filename | Format & Schema | Consuming Core Function | Handshake Role |
|---|---|---|---|
| `frame_0.png` | 300 DPI Transparent PNG | `class-util.php:2673 explode('_', $arr[0])[1]` | **Critical:** Side 0 print raster. NBDesigner parses the side index from the filename. |
| `preview/frame_0.png` | Garment Mockup PNG | `box-order-metadata.php :41 get_list_images(..., 1)` | **Critical:** Thumbnail displayed in WooCommerce Order 'Customer Design' metabox. |
| `svg/frame_0.svg` | Vector SVG scene-graph | `class.nbdesigner.php:1 975 download-type=svg` | **Critical:** Packaged into customer-design-...-svg.zip for vector cutters. |

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| `frame_0.svg`<br>`design.svg` | Vector SVG scene-graph | `convert_svg_embed()` | Root vector assets for external ERPs and conversion fallback. |
| `design.json` | `Fabric / NBDesigner JSON {'frame_0': {objects: [...]}}` | `detail-order.php:143 foreach($resources as $res)` | **Critical:** Populates Image Layers and Text Layers tables in 'View Detail'. |
| `product.json` | Serialized PHP array of product sides | `class.nbdesigner.php:2040 nbd_export_pdfs()` | Defines physical width, height, bleed, and orientation dimensions. |
| `option.json` | `Serialized PHP array [dpi=>300, unit=>'cm', layout=>'m']` | `detail-order.php:8 (float)$option['dpi']` | Supplies resolution and measurement unit calculations. |
| `config.json` | `JSON metadata container {dpi: 300, unit: 'cm', product: [...]}` | `class.nbdesigner.php:1942` | Canvas bounding box metadata and original payload snapshot. |
| `used_font.json` | JSON array (safe: []) | `class-util.php:2623 nbd_download_google_font()` | **Critical:** Must be valid JSON array. An empty array avoids TCPDF font download fatal errors. |
| `print_file_front.png`<br>`print_300dpi.png` | 300 DPI Transparent PNG | `OrderAdminPanel.php` | Direct high-res print download aliases for quick access. |

# 3. NBDesigner Core Code Deep-Dive: Why Each File is Mandatory

## A. Order Metabox Rendering Gate (class.nbdesigner.php:2244 & 2256)

NBDesigner registers an order metabox (views/box-order-metadata.php). However, it contains an explicit gating check:

```
$has_design = get_post_meta( $order_id, '_nbd', true );
$has_upload = get_post_meta( $order_id, '_nbu', true );
if( $has_design || $has_upload ) {
    include_once( NBDESIGNER_PLUGIN_DIR . 'views/box-order-metadata.php' );
}
```

**Rule:** If postmeta _nbd is not set to 1 on the WooCommerce order, the entire metabox is hidden. The bridge must execute both $order->update_meta_data('_nbd', '1') (for HPOS) and update_post_meta($order_id, '_nbd', '1').

## B. The preview/ Subfolder Thumbnail Requirement (views/box-order-metadata.php:41)

Inside the metabox, NBDesigner renders the customer's design preview using:

```
$list_images = Nbdesigner_IO::get_list_images( NBDESIGNER_CUSTOMER_DIR . '/' . $nbd_item_key . '/previ
ew', 1 );
asort( $list_images );
if( count( $list_images ) > 0 ):
    foreach( $list_images as $key => $image ):
        $src = Nbdesigner_IO::convert_path_to_url( $image );
        echo '<img class="nbdesigner_order_image_design" src="' . esc_url($src) . '" />';
```

**Rule:** Storing design_preview.png only in the root directory leaves $list_images empty. The file **must** exist at preview/frame_0.png.

## C. Filename Explode for Vector PDF Export (includes/class-util.php:2673)

When generating print-ready PDFs via TCPDF, NBDesigner iterates through root images:

```
$list_images = Nbdesigner_IO::get_list_images( $path, 1 );
foreach ( $list_images as $img ){
    $name               = basename( $img );
    $arr                = explode( '.', $name );
    $_frame             = explode( '_', $arr[0] );
    $frame              = $_frame[1];             // Must yield '0' for Front side!
    $list_design[$frame] = $img;
}
```

**Rule:** If named print_file_front.png, $_frame[1] evaluates to 'file', meaning $list_design[0] remains empty and the exported PDF page contains no graphics! The file **must be named frame_0.png**.

## 4. Universal Drop-In PHP Handshake Adapter

The following self-contained class can be included in any custom WordPress plugin or theme. Call UniversalNBDBridge::inject_nbd_artifacts(...) during Add-to-Cart or Checkout to establish the complete 10-artifact handshake:

```
class UniversalNBDBridge {
    public static function inject_nbd_artifacts( $nbd_key, $custom_data, $preview_b64, $svg_markup, $p
rint_b64, $product_id ) {
        $upload  = wp_upload_dir();
        $nbd_dir = trailingslashit( $upload['basedir'] ) . 'nbdesigner/designs/' . sanitize_file_name(
 $nbd_key );
        $nbd_url = trailingslashit( $upload['baseurl'] ) . 'nbdesigner/designs/' . sanitize_file_name(
 $nbd_key );
        wp_mkdir_p( $nbd_dir . '/preview' );
        wp_mkdir_p( $nbd_dir . '/svg' );

        // 1. Preview images
        if ( ! empty( $preview_b64 ) ) {
            $bin = base64_decode( strpos($preview_b64, 'base64,') !== false ? explode('base64,', $prev
iew_b64)[1] : $preview_b64 );
            file_put_contents( $nbd_dir . '/design_preview.png', $bin );
            file_put_contents( $nbd_dir . '/preview/frame_0.png', $bin );
        }
        // 2. Vector SVG scene-graphs
        if ( ! empty( $svg_markup ) ) {
            file_put_contents( $nbd_dir . '/design.svg', $svg_markup );
            file_put_contents( $nbd_dir . '/frame_0.svg', $svg_markup );
            file_put_contents( $nbd_dir . '/svg/frame_0.svg', $svg_markup );
        }
        // 3. 300 DPI Transparent print rasters
        if ( ! empty( $print_b64 ) ) {
            $bin = base64_decode( strpos($print_b64, 'base64,') !== false ? explode('base64,', $print_
b64)[1] : $print_b64 );
            file_put_contents( $nbd_dir . '/frame_0.png', $bin );
            file_put_contents( $nbd_dir . '/print_file_front.png', $bin );
            file_put_contents( $nbd_dir . '/print_300dpi.png', $bin );
        }
        // 4. Product sides configuration (cm)
        $sides = array( array(
            'orientation_name' => 'Front', 'product_width' => 30, 'product_height' => 40,
            'real_width' => 20, 'real_height' => 26, 'real_top' => 5, 'real_left' => 5,
            'area_design_width' => 2400, 'area_design_height' => 3000, 'img_src' => '',
            'bg_type' => 'color', 'bg_color_value' => '#ffffff', 'bleed_top_bottom' => 0,
            'bleed_left_right' => 0, 'include_background' => 0
        ) );
        file_put_contents( $nbd_dir . '/product.json', serialize( $sides ) );
        file_put_contents( $nbd_dir . '/option.json', serialize( array('dpi' => 300, 'unit' => 'cm', '
layout' => 'm') ) );
        file_put_contents( $nbd_dir . '/config.json', json_encode( array('dpi' => 300, 'unit' => 'cm',
 'product' => $sides) ) );
        file_put_contents( $nbd_dir . '/used_font.json', '[]' );

        // 5. design.json (Fabric objects for 'View Detail')
        file_put_contents( $nbd_dir . '/design.json', json_encode( array('frame_0' => array('version'
=> '3.4.0', 'objects' => $custom_data['objects'])) ) );
    }
}
```

## 5. Live Production Verification Checklist

- **Order Postmeta Check:** Run wp post meta get {order_id} _nbd. Must return 1.
- **Line Item Meta Check:** Ensure order item meta contains _nbd = {nbd_item_key} and _nbd_cart_item_key = {nbd_item_key}.
- **Metabox Visibility:** Navigate to wp-admin/post.php?post={order_id}&action;=edit. The 'Customer Design' box displays with thumbnail, checkbox, and 'View detail' link.
- **1-Click Detail View:** Click 'View detail'. The Resource tab lists all text lines, colors, fonts, and graphical elements.
- **Vector PDF Export:** Click 'Create PDF' or trigger nbd_export_pdfs('{nbd_item_key}'). Generates customer-pdfs/{nbd_item_key}_0.pdf cleanly.
- **SVG Archive Export:** Click 'Download SVG'. Downloads customer-design-{nbd_item_key}-svg.zip containing svg/frame_0.svg.